8 Information Theory & Coding

Objective

After completing this chapter, user will be able to understand:

1. Basics of information theory.
2. What entropy is and its property.
3. Different types of channels.
4. Mutual information.
5. Source coding, channel coding theorem.
6. Linear code.

Introduction

1. The main work of a communication system is to transmit information from an information source (i.e., transmitter) to a destination via a communication channel.
2. When we send a message from transmitter to receiver, then, we actually send information from transmitter to receiver.
3. In this chapter, we are going to study communication channels and the concept of coding for the protection of messages against channel error.

Entropy and Information

In a communication system, the source uses the transmitter to send information. Let a source be a discrete and memory less source. Here, 'discrete' means that the source sends one symbol at a time and 'memoryless' means that the current symbol is independent of previous symbols. Let the source produce symbols.

o $\{S_0, S_1, S_2, \ldots S_N-1\}$

And, probability of symbols is P_k

Now, $\sum_{k=0}^{k-1} P_k = 1 \left(\begin{array}{l} sum\,of\,all \\ probability \end{array} \right)$

Information content in a symbol is inversely proportional to probability

$$i.e,\ l_k \alpha \frac{1}{P_k}$$

$$l_k = \log_2 \left(\frac{1}{probability\ of\ symbol} \right)$$

$$l_k = \log_2 \frac{1}{P_k}$$

Less probability means more uncertainty and more information.

$I(S_k) = 0$, if, probability of any symbol = 1

$I(S_k) > I(S_i)$, if, probability of $S_k <$ probability of S_i

And, $I(S_k S_i) = \log_2 \frac{1}{P_k P_i}$

$$= \log_2 \frac{1}{P_k} + \log_2 \frac{1}{P_1}$$

$$= I(S_k) + I(S_i)$$

$I(S_k . S_i)$ = Information is transferred when S_k and S_i occur together or are received together by the receiver.

$$= \log_2 \frac{1}{Probability\ of\ S_k\ \&\ S_i\ occur\ together}$$

$$= \log_2 \left(\frac{1}{P_{s_i} . P_{s_k}} \right)$$

The unit of information is 'bits'.

Example:

Calculate the in formation in a symbol if its probability is $\frac{1}{16}$.

Solution:

Given,

Probability of symbol = $\frac{1}{16}$

We know that,

Information of symbol

$$= \log_2 \left(\frac{1}{Probability\ of\ symbol} \right)$$

$$= \log_2 \left(\frac{1}{1/16} \right)$$

$$= \log_2 16$$

= 4 bits

Hence, the symbol contains 4 bits of information.

Entropy of the Source

The entropy of the source is defined as the expected value of information transmitted by the source.

$$E[I(S_k)] = \sum_{k=0}^{k-1} I(S_k).P_k$$

$$E\left[I(S_k)\right] = \sum_{k=0}^{k-1} P_k \log_2 \frac{1}{P_k} \; Bits / symbol$$

From the above expression, we can say that entropy gives us the average information per symbol.

Unit of entropy ⇒ Bits/symbol.

Example:

Find the entropy of a source that emits one of the three symbols x, y, z in a statistically independent sequence with probabilities

$\frac{1}{4}, \frac{1}{8}$ *and* $\frac{5}{8}$ respectively.

Solution:

Entropy of symbol = $E[I(S_k)]$

$$= \sum_{k=0}^{k-1} P_k \log_2 \left(\frac{1}{P_k}\right)$$

$$= \frac{1}{4}\log_2\left(\frac{1}{1/4}\right) + \frac{1}{8}\log_2\left(\frac{1}{1/8}\right) + \frac{5}{8}\log_2\left(\frac{1}{5/8}\right)$$

$$= \frac{1}{4}\log_2(4) + \frac{1}{8}\log_2 8 + \frac{5}{8}\log_2\frac{8}{5}$$

$$= \frac{1}{2} + \frac{3}{8} + 0.424$$

= 1.299 bits per symbol

Information Rate

Information rate is the rate, at which, information is passed over the channel.

Information rate = symbol rate × entropy of the source

$R = r_s \times E$

R = Information rate

r_s = Symbol rate

E = Entropy of the source

Example:

A discrete source emits one symbol, once in every microsecond. The symbol probabilities are $\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ respectively. Find the source entropy and information rate.

Solution:

$$\text{Source entropy} = \sum_{k=0}^{k-1} P_k \log_2 \frac{1}{P_k}$$

Source entropy

$$= \frac{1}{4}\log_2\left(\frac{1}{1/4}\right) + \frac{1}{4}\log_2\left(\frac{1}{1/4}\right)$$

$$+ \frac{1}{4}\log_2\left(\frac{1}{1/4}\right) + \frac{1}{8}\log_2\left(\frac{1}{1/8}\right) + \frac{1}{8}\log_2\left(\frac{1}{1/8}\right)$$

$$= \frac{3}{4}\log_2(4) + \frac{2}{8}\log_2(8)$$

$$\text{Source entropy} = \frac{3}{2} + \frac{3}{4}$$

= 2.25 bits per symbols

Information rate R = Symbol rate × Entropy

Symbol rate = 10^6 symbol/sec

Information rate = 2.25 × 10^6 bits/sec

Special case:

If we have k symbols, then the entropy of the source is maximum when the symbols are equiprobable; *i.e*, $Ps_k = \frac{1}{k}$

$$\text{Entropy} = \sum_{k=0}^{k-1} P_k \log \frac{1}{P_k}$$

$$= \frac{1}{k}\log_2 k + \frac{1}{k}\log_2 k \ldots k \; times = \frac{1}{k} \times k \log_2 k$$

Entropy = $\log_2 k$ ⇒ In case of equiprobable symbols, entropy is maximum = $\log_2 k$

Where, k = Number of symbols.

Minimum entropy = 0

(When, any 1 symbol is with probability = 1)

Hence, range of entropy:

0 ≤ entropy of the source ≤ $\log_2 k$.

Example:

Calculate the entropy of the source, if, 4 symbols are transmitted with probability 0.4,

0.3, 0.2 and 0.1 respectively and also calculate the entropy, if, symbols are equiprobable.

Solution:

Entropy of the source

$$\sum_{k=0}^{k-1} P_k \log_2 \frac{1}{P_k}$$

$$= 0.4\log_2\left(\frac{1}{0.4}\right) + 0.3\log_2\left(\frac{1}{0.3}\right) + 0.2\log_2\left(\frac{1}{0.2}\right) + 0.1\log_2\left(\frac{1}{0.1}\right)$$

= 0.5288 + 0.5210 + 0.4644 + 0.3321

= 1.8463 bits/symbol

Now, if symbols are equiprobable,

Entropy = $\log_2 k$

K = Number of symbols = 4

Entropy = $\log_2 k$

= $\log_2$ 4 = 2 bits/symbol

Entropy of source, if, all symbols are equiprobable

= 2 bits/symbol

Extension of discrete memoryless source:

Suppose, we have a source with an entropy of H(x), and we create an extended source, where 'n' such sources are connected to produce an extended source. The entropy of the extended source is

$H(x^n) = nH(x)$

= n × entropy of original source

e.g. Let a source produce 2 symbols (S_0, S_1) with a probability of 0.3 and 0.7 respectively. Now, if we extend the source and design, in such a way that it produces 2 symbols at a time.

Then, extended source output symbols are as $S_0S_0 \rightarrow 0.3 \times 0.3 = 0.09$

$S_0S_1 \rightarrow 0.3 \times 0.7 = 0.21$

$S_1S_0 \rightarrow 0.7 \times 0.3 = 0.21$

$S_1 S_1 \rightarrow 0.7 \times 0.7 = 0.49$

Entropy of the source = $\sum_{k=0}^{k-1} P_k \log_2 \frac{1}{P_k}$

Entropy of the original source = $0.3 \log_2 \left(\frac{1}{0.3}\right) + 0.7\log_2\left(\frac{1}{0.7}\right)$

= 0.5211 + 0.3602

= 0.8813 bits/symbol

And,

Entropy of the extended source

$$= 0.09 \log_2\left(\frac{1}{0.09}\right) + 0.21\log_2\left(\frac{1}{0.21}\right) + +0.21\log_2\left(\frac{1}{0.21}\right) + 0.49\log_2\left(\frac{1}{0.49}\right)$$

= 0.3127 + 0.4728 + 0.4728 + 0.5043

= 1.7626 bits/symbol

= 2 × Entropy of original source.

Condition Probability

Condition Probability is a measure of the probability of an event occurring, given that another event has already occurred. But, here, we will discuss about condition probability for a channel.

Assume that a discrete memoryless source produces 'm' symbols, e.g., (x_1, x_2 ...x_m) and transmits them through a channel. Let the received symbol at the receiver be as (y_1, y_2 ... y_m).

Now,

$P_{Y_j}\left(\frac{Y_j}{X_k}\right)$ = Represents the probability of getting

Given that x_k is transmitted.

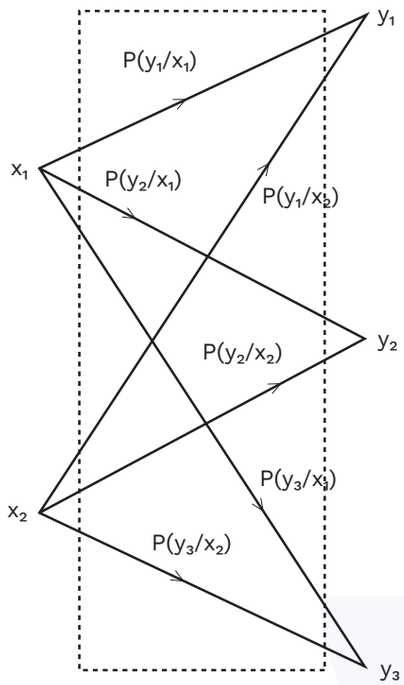
The channel matrix or channel conditional matrix

$$P\left(\frac{Y}{X}\right) = \begin{bmatrix} P\left(\frac{Y_1}{X_1}\right) & P\left(\frac{Y_2}{X_1}\right) & \ldots P\left(\frac{Y_n}{X_1}\right) \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ P\left(\frac{Y_1}{X_m}\right) & P\left(\frac{Y_2}{X_m}\right) & P\left(\frac{Y_n}{X_m}\right) \end{bmatrix}$$

Now,

Let's assume, a source is transmitting 2 symbols (x_1 and x_2) through a channel.

At receiver's end, the received symbols are y_1, y_2, y_3



Channel

i.e.

$$P(y_1) = P(x_1)P\left(\frac{y_1}{x_1}\right) + P(x_2)P\left(\frac{y_1}{x_2}\right)$$

$$P(y_2) = P(x_1)P\left(\frac{y_2}{x_1}\right) + P(x_2)P\left(\frac{y_2}{x_2}\right)$$

$$P(y_3) = P(x_1)P\left(\frac{y_3}{x_1}\right) + P(x_2)P\left(\frac{y_3}{x_2}\right)$$

$$P(y) = P(x)\left(\frac{y}{x}\right)$$

i.e.

$$\underbrace{[P(y_1)\ P(y_2)\ P(y_3)]}_{\text{Output symbol probability}} = \underbrace{[P(x_1)P(x_2)]}_{\text{Input symbol probability}} \underbrace{\begin{bmatrix} P\left(\frac{y_1}{x_1}\right) & P\left(\frac{y_2}{x_1}\right) & P\left(\frac{y_3}{x_1}\right) \\ P\left(\frac{y_1}{x_2}\right) & P\left(\frac{y_2}{x_2}\right) & P\left(\frac{y_3}{x_2}\right) \end{bmatrix}}_{\text{Channel matrix}}$$

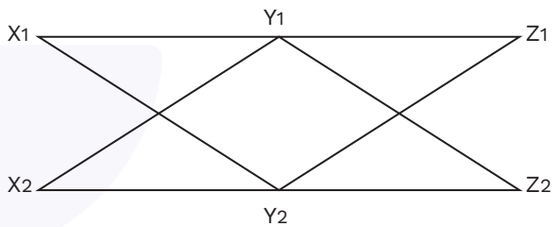
Properties of channel matrix:

$$P(y) = [P(x_1)\ P(x_2)\ldots P(x_m)]\left[P\left(\frac{Y}{X}\right)\right]$$

1. Sum of elements of any row of channel matrix = 1
2. $i.e. \sum P\left(\frac{y_i}{x_i}\right) = 1$

This is because, when x_i is transmitted, we surely get an output. (Any one form y_1, y_2 ...y_n)

3. Number of columns in channel matrix shows the number of output symbols.
4. Number of rows in channel matrix shows number of input symbols.
5. If, multiple channels are in cascade, then the channel matrix for the whole channel is the product of all channel's matrix. E.g.

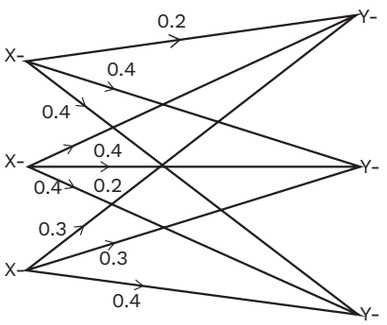


$$P\left(\frac{Z}{X}\right) = P\left(\frac{Y}{X}\right)P\left(\frac{Z}{Y}\right)$$

$$P\left(\frac{Z}{X}\right) = \begin{bmatrix} P\left(\frac{Y_1}{X_1}\right)P\left(\frac{Y_2}{X_1}\right) \\ P\left(\frac{Y_1}{X_2}\right)P\left(\frac{Y_2}{X_2}\right) \end{bmatrix} \begin{bmatrix} P\left(\frac{Z_1}{Y_1}\right)P\left(\frac{Z_2}{Y_1}\right) \\ P\left(\frac{Z_1}{Y_2}\right)P\left(\frac{Z_2}{Y_2}\right) \end{bmatrix}$$

Example:

Find the output symbol probability for the given channel with $P(x_1) = 0.2$, $P(x_2) = 0.4$ and $P(x_3) = 0.4$.



Solution:

Given

$P(x_1) = 0.2$, $P(x_2) = 0.4$, $P(x_3) = 0.4$

For the given channel

$$P\left(\frac{Y_1}{X_1}\right) 0.2, P\left(\frac{Y_2}{X_1}\right) = 0.4\, P\left(\frac{Y_3}{X_1}\right) = 0.4$$

$$P\left(\frac{Y_1}{X_2}\right) 0.3, P\left(\frac{Y_2}{X_2}\right) = 0.2\, P\left(\frac{Y_3}{X_2}\right) = 0.4$$

$$P\left(\frac{Y_1}{X_3}\right) 0.3, P\left(\frac{Y_2}{X_3}\right) = 0.3\, P\left(\frac{Y_3}{X_3}\right) = 0.4$$

We know that

$$P[Y] = P[X]\, P\left[\frac{Y}{X}\right]$$

$$= P[X] = [0.2\,0.4\,0.4]$$

Channel matrix,

$$P\left[\frac{Y}{X}\right] = \begin{bmatrix} 0.2 & 0.4 & 0.4 \\ 0.4 & 0.2 & 0.4 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

$$P[y] = [0.2\ 0.4\ 0.4] \begin{bmatrix} 0.2 & 0.4 & 0.4 \\ 0.4 & 0.2 & 0.4 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

= [0.04+0.16+0.120.08+0.08+0.12 08+0.16+0.16]

$P(y_1) = 0.32$ $P(y_2) = 0.28$ $P(y_3) = 0.4$

Example:

Given, a channel matrix

$$P\left(\frac{y}{x}\right) = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.4 & 0.3 \end{bmatrix}$$

Find out the number of input and number of output symbols for the given channel matrix. Also, is this a valid channel matrix?

Solution:

Number of inputs = Number of rows = 3

Number of outputs = Number of columns = 3

For valid channel matrix, sum of elements of any row is = 1

Here, Row 1 $\Rightarrow$ 0.2 + 0.3 + 0.5 = 1

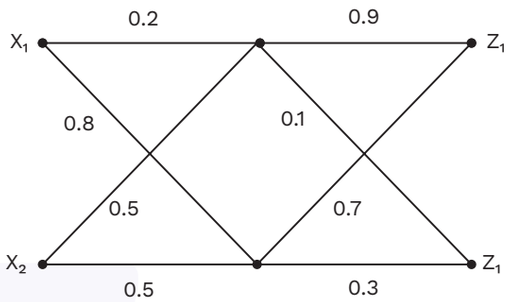
Row 2 $\Rightarrow$ 0.1 + 0.6 + 0.3 = 1

Row 3 $\Rightarrow$ 0.2 + 0.4 + 0.3 = 0.9

As sum of element of 3rd row is not equal to 1. Hence, it is not a valid channel matrix.

Example:

Two channels A and B are in cascade as shown in the figure



Find the output symbols' probability, if, input symbols x_1 and x_2 are equiprobable.

Solution:

Given,

$$P(x_1) = P(x_2) = 0.5$$

For the given channel

$$P\left[\frac{Y}{X}\right] = \begin{bmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{bmatrix}$$

$$P\left[\frac{Z}{Y}\right] = \begin{bmatrix} 0.9 & 0.1 \\ 0.7 & 0.3 \end{bmatrix}$$

Complete channel matrix = Product of both channels' matrix.

$$P\left[\frac{Z}{X}\right] = P\left(\frac{Y}{X}\right) P\left(\frac{Z}{Y}\right)$$

$$P\left(\frac{Z}{X}\right) = \begin{bmatrix} 0.2 & 0.8 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.7 & 0.3 \end{bmatrix}$$

$$P\left(\frac{Z}{X}\right) = \begin{bmatrix} 0.18+0.56 & 0.02+0.24 \\ 0.45+0.35 & 0.05+0.15 \end{bmatrix}$$

$$P\left(\frac{Z}{X}\right) = \begin{bmatrix} 0.74 & 0.26 \\ 0.8 & 0.2 \end{bmatrix}$$

Now,

$$P(z)=P(x)P\left(\frac{z}{x}\right)$$

$$=\begin{bmatrix}0.5 & 0.5\end{bmatrix}\begin{bmatrix}0.74 & 0.26\\ 0.8 & 0.2\end{bmatrix}$$

= [0.37 + 0.4 0.13+0.1]

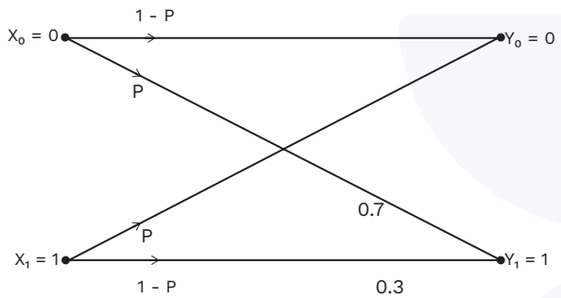
P(z) = [0.77 0.23]

From P(z) matrix

$$\left.\begin{matrix}P(Z_1)=0.77\\ P(Z_2)=0.23\end{matrix}\right\}\ \textit{output symbol probabilities}$$

Binary Symmetric Channel (BSC)

This type of channel transmits only two distinct characters, generally interpreted as 0 and 1. In a binary symmetric channel, there are 2 inputs and 2 outputs as shown below.



P = Crossover probability

Channel matrix for BSC $=\begin{bmatrix}1-P & P\\ P & 1-P\end{bmatrix}$

Cascading of two BSC's:

$$=\begin{bmatrix}(1-P)^2+P^2 & 2P(1-P)\\ 2P(1-P) & (1-P)^2+P^2\end{bmatrix}$$

$$P\left(\frac{x}{y}\right)=\begin{bmatrix}1-P & P\\ p & 1-P\end{bmatrix}\begin{bmatrix}1-P & P\\ p & 1-P\end{bmatrix}$$

$$\boxed{H\left(\frac{Y}{z}\right)=P\log_2 P+(1-P)\log_2(1-P)}\rightarrow H(P)$$

$$\Rightarrow\begin{bmatrix}1-2P+2P^2 & 2P(1-P)\\ 2P(1-P) & 1-2P+2P^2\end{bmatrix}$$

$$\Rightarrow\begin{bmatrix}1-2P(1-P) & 2P(1-P)\\ 2P(1-P) & 2P(1-P)\end{bmatrix}$$

= [(1 – 2P(1 – P)$\log_2$ (1 –2P(1 –P)]+ 2P(1 – P) $\log_2$P(1 – P)]

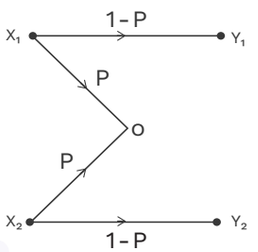
$\Rightarrow$ H[2P(1 – P)]

P = Crossover probability

$$\boxed{\begin{matrix}H\left(\frac{z}{x}\right)\geq H\left(\frac{y}{x}\right)\\ I(x,z)\leq I(x,y)\end{matrix}}$$

Binary Erasure Channel (BEC)

Binary Erasure Channel (BEC)



$$P\left(\frac{y}{z}\right)=\begin{bmatrix}1-P & P & O\\ O & P & 1-p\end{bmatrix}$$

Where, p → Erasure probability (x_1 is transmitted as y_1 but not as y_2; data is deleted in this and due to this, it is known as Erasure channel).

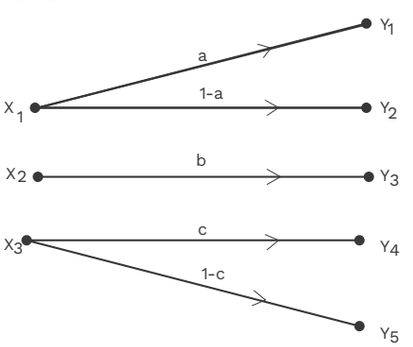
$$I(x,y)=H(x)-H\left(\frac{x}{y}\right)$$

$$I(x,y)=(1-p)H(x)\quad \because \max H(x)=\log_2 2=1$$

$$\boxed{C=\max I(x,y)=(1-p)}$$

Lossless Channel

The channel is called lossless if by looking at the output, we can find out which input is transmitted.



Here, we can find out the input, easily, by output.

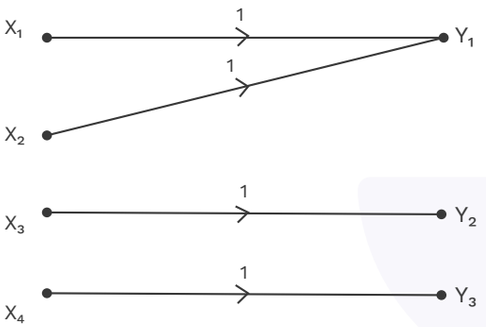
$$P\left(\frac{y}{x}\right)=\begin{bmatrix} a & 1-a & 0 & 0 & 0 \\ 0 & 0 & b & 0 & 0 \\ 0 & 0 & 0 & c & 1-c \end{bmatrix}$$

In lossless channel matrix, only one element of each column is non-zero.

Deterministic Channel

In deterministic channel, if we know the input, then we can, surely, determine the output.

E.g.

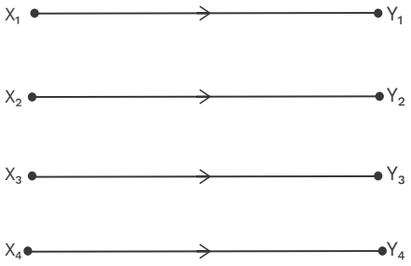


Channel matrix, $P\left(\frac{y}{x}\right)=\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

In deterministic channel matrix, only one element of each row is non-zero.

Noiseless Channel

If channel is both loss-less and deterministic, i.e., looking at input, we can find output and looking at output, we can find out input.



Channel matrix = $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Mutual Information

Suppose that we have a communication system where input to channel is x and output from channel is y. Let there be 'n' input symbols and 'm' output symbols.



Information $I(X_K)=\log_2\frac{1}{P_{x_k}}$

And, entropy $H(x)=\sum_{K=0}^{N-1}P_{X_k}\log_2\frac{1}{P_{X_k}}$

Information tells about the amount of uncertainty, while, entropy tells about average uncertainty in input symbols.

$H\left(\frac{x}{y=y_k}\right)$

= Uncertainty in x and given that y_k is received

$\sum_{i=0}^{N-1}P\left[\frac{X_i}{Y_k}\right]\log_2\frac{1}{P(x_i/y_k)}$

$H\left[\frac{X}{Y}\right]$ = Mean of H $\left[\frac{x}{y=y_k}\right]$

$H\left(\frac{x}{y}\right)=\sum_{k=0}^{N-1}P_{y_k}H\left(\frac{x}{y=y_k}\right)$

$=\sum_{k=0}^{M-1}\sum_{i=0}^{N-1}P\left(\frac{x_i}{y_k}\right)\log_2\frac{1}{P(x_i/y_k)}$

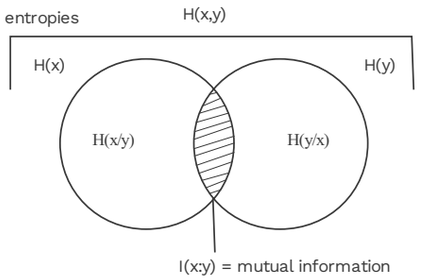
$H\left(\frac{X}{Y}\right)=\sum_{k=0}^{M-1}\sum_{i=0}^{N-1}P\left(\frac{x_i}{y_k}\right)\log_2\frac{1}{P(x_i/y_k)}$

$H\left(\frac{X}{Y}\right)$ represents the amount of uncertainty in input after receiving the output.

H(x) = Entropy of output.

Information transferred = H(x) – H(x/y) = I(x:y) and this is also called mutual information.

⇒ Venn diagram representation of various channels.



From Venn diagram:

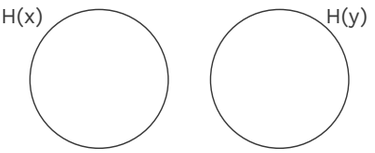
1. Mutual information = Intersection b/w H(x) and H(y)
2. Mutual information = x H(x) H $\left(\frac{x}{y}\right)$
   $= H(y) - H\left(\frac{y}{x}\right)$
3. If noise = 0, then, information transferred = H(x)
4. $H(x) = \sum_{i=0}^{N-1} P_{x_i} \log_2 \frac{1}{P_{x_i}}$
   $H(y) = \sum_{j=0}^{M-1} \sum_{j=0}^{N-1} P(X_i, y_j) \log_2 \frac{1}{P_{x_i}}$
5. Mutual information = x H(x) H $\left(\frac{x}{y}\right)$
   $I(x;y) = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} P(x_i, y_j) \log_2 \frac{P(x_i / y_j)}{P(x_i)}$
6. Mutual information, (x : y) = H $\left(\frac{x}{y}\right)$
   $= \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} P(x_i, y_j) \log_2 \frac{P(x_i / y_j)}{P(x_i)}$
7. Total Venn diagram
   $I(x;y) + H\left(\frac{y}{x}\right) + H\left(\frac{x}{y}\right)$

$$= \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} P(x_i, y_j) \left[\log_2 \frac{P(x_i / y_j)}{p(x_i)} + \log_2 \frac{1}{P(x_i / y_j)} + \log_2 \frac{1}{P(y_j / x_i)} \right]$$

Joint entropy

$$H(x_i, y_j) = \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} P(x_i, y_j) \left[\log_2 \frac{1}{P(x_i / y_j)} \right]$$

8. 8. Mutual information I(x : y) = H(x)+H(y)-H(x,y)
9. 9. Mutual information is never negative i.e. I(x ; y) ≥ 0 always
10. 10. I(x ; y) = 0, when x and y are independent, then Venn diagram



I(x : y) = 0

Example:

Let, 'A' represents whether it is sunny or rainy in a particular town on a given day. Let, 'y' represents whether temperature is above 30°C or below 30°C. Compute the entropy of the joint distribution. P(x, y) given by

P(sunny, below 30°C) = $\frac{1}{2}$

P(sunny, above 30°C) = $\frac{1}{4}$

P(sunny, below 30°C) = $\frac{1}{8}$

P(sunny, above 30°C) = $\frac{1}{8}$

Solution:

Joint entropy

$$H(x, y) = \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} P(x_i, y_j) \left[\frac{1}{P(x_i, y_j)} \right]$$

$$H(x, y) = \frac{1}{2} \log_2 \left(\frac{1}{1/2} \right) + \frac{1}{4} \log_2 \left(\frac{1}{1/4} \right) + \frac{1}{8} \log_2 \left(\frac{1}{1/8} \right) + \frac{1}{8} \log_2 \left(\frac{1}{1/8} \right)$$

$$= \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{1}{8} \log_2 8 + \frac{1}{8} \log_2 8$$

$$= \frac{1}{2} + \frac{1}{4} \times 2 \frac{2}{8} \times 3$$

H(x, y) = 1.75

Example:

Two binary random variables x and y are distributed according to the joint PMF given by

$P(x = 0, y = 1) = \frac{1}{4}$

$P(x = 1, y = 1) = \frac{1}{4}$

$P(x = 1, y = 0) = \frac{1}{8}$

$P(x = 0, y = 0) = \frac{3}{8}$

Determine H(x, y), H(x), H(y), H(x/y), H(y/x) and I(x ; y).

Solution:

$$H(x,y)=\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}P(x_i,y_j)\log_2\frac{1}{P(x_i,y_j)}$$

$$=\frac{1}{4}\log_2\frac{1}{1/4}+\frac{1}{4}\log_2\frac{1}{1/4}$$

$$+\frac{1}{8}\log_2\frac{1}{1/8}+\frac{3}{8}\log_2\frac{1}{3/8}$$

$$=\frac{1}{4}\log_2 4+\frac{1}{4}\log_2 4+\frac{1}{8}\log_2 8+\frac{3}{8}\log_2\frac{3}{8}$$

$$=1+\frac{3}{8}+0.531$$

H(x, y) 1.906 = Bits per symbol

P(x=1) = P(x=1, y=0) + P(x=1, y=1)

$$=\frac{1}{4}+\frac{1}{8}$$

$$=\frac{3}{8}$$

P(x=0) = P(x=0, y=0) + P(x=0, y=1)

$$=\frac{3}{8}+\frac{1}{4}$$

$$=\frac{5}{8}$$

$$H(x)=\frac{3}{8}\log_2\frac{1}{3/8}+\frac{5}{8}\log_2\frac{1}{5/8}$$

$$=\frac{3}{8}\log_2\frac{8}{3}+\frac{5}{8}\log_2\frac{8}{5}$$

= 0.531 + 0.4238

H(x) 0.9548 = bits per symbol

P(y = 0) = P(x = 0, y = 0) + P(x = 1, y = 0)

$$=\frac{3}{8}+\frac{1}{8}$$

$$=\frac{4}{8}=\frac{1}{2}$$

$P(y = 0) = \frac{1}{2}$

P(y = 1) = P(x = 0, y = 1) + P(x = 1, y = 1)

$$=\frac{1}{4}+\frac{1}{4}=\frac{1}{2}$$

$$P(y=1)=\frac{1}{2}$$

$$H(y)=\frac{1}{2}\log_2\frac{1}{1/2}+\frac{1}{2}\log_2\frac{1}{1/2}$$

$$=\frac{1}{2}\log_2\frac{1}{2}\log_2$$

H(y) 1 = bits per symbol

$H(x, y) = H(x) + y\ H\left(\frac{y}{x}\right)$

$H\left(\frac{y}{x}\right) = H(x, y) - H(x)$

$H\left(\frac{y}{x}\right) = 1.906 - 0.9548$

$H\left(\frac{y}{x}\right) = 0.9512$ bits per symbol

$H(x, y) = H(y) + H\left(\frac{x}{y}\right)$

$H\left(\frac{x}{y}\right) = H(x, y) - H(y)$

$H\left(\frac{x}{y}\right) = 1.906 - 1$

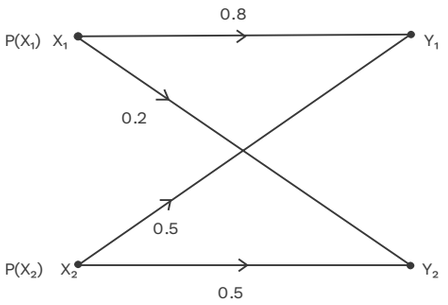
$H\left(\frac{x}{y}\right) = 0.906$

$I(x, y) = H(x) - H\left(\frac{x}{y}\right) = 0.9548 - 0.906$

I(x, y) = 0.0488

Example:

Consider a binary channel shown in the figure



Find the joint probabilities $P(x_1, y_2)$ and $P(x_2, y_1)$ when $P(x_1) = P(x_2) = 0.5$.

Solution:

$$[P(x, y)] = [P(x)]_d \left[P\left(\frac{y}{x} \right) \right]$$

$$[P(x)]_d = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$$[P(x, y)] = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.5 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4 & 0.1 \\ 0.25 & 0.25 \end{bmatrix} = \begin{bmatrix} P(x_1, y_1) & P(x_1, y_2) \\ P(x_2, y_1) & P(x_2, y_2) \end{bmatrix}$$

Therefore,

$$P(x_1, y_2) = 0.1$$

$P(x_2, y_1) = 0.25$

Shannon Hartley Law

Shannon Hartley Law relates the system capacity of a channel with the average received signal power, the average noise power and the bandwidth.

$$C = B \log \left(1 + \frac{S}{N} \right)$$

C = Capacity of the channel (bits/s)

S = Average receive signal power

N = Average noise power

B = Bandwidth (Hz)

Channel Capacity

The channel capacity, C, is defined as the maximum rate at which information can be transmitted through a channel.

Mutual Information:

$$I(X;Y) = \left[\sum_{K=0}^{M-1} \sum_{i=0}^{N-1} P(x_i, y_k) \log_2 \frac{P(x_i / y_k)}{P(x_i)} \right]$$

Mutual information that is being transmitted is dependent on channel or on the channel matrix $P\left[\frac{y}{x} \right]$ and also, depends on the source i.e., P[x].

The channel capacity is also defined as the maximum mutual information in any single transmission.

$\Rightarrow$ Channel capacity = Max I[x ; y]

$$I[x ; y] = H(x) - H\left(\frac{y}{x} \right)$$

$\Rightarrow$ Channel capacity = Max. I[x ; y] bits/symbol

or

Channel capacity = R × (max. I(x ; y)) bits/sec, where, symbol rate = R symbol/sec y. For finding the maximum information, we assume source to be equiprobable.

Channel capacity of lossless channel:

For lossless channel, $H\left(\frac{x}{y} \right) = 0$

$$I[X;Y] = H(X) - H\left(\frac{x}{y} \right) \left(H\left[\frac{x}{y} \right] = 0 \right)$$

= H[x]

Capacity = Max I(x ; y) = Max H(x)

$= \log_2 N$

N = Number of symbols in source input

Channel capacity of deterministic channel:

For deterministic channel, $H\left[\frac{y}{x} \right] = 0$

I[x : y] = H[y] – 0

= H[y]

Channel capacity = Max I[x ; y] = Max H[y] = $\log_2 M$; M is no. of symbols in output.

Channel capacity of noiseless channel:

For noiseless channel

$$H\left(\frac{y}{x}\right)=H\left(\frac{x}{y}\right)=0$$

$$I[x;y]=H[x]-H\left(\frac{x}{y}\right)=H(x)$$

$$I[x;y]=H[y]-H\left(\frac{y}{x}\right)$$

$$H(y)$$

I[x ; y] = H(x) = H(y)

Channel capacity = Max I(x ; y)

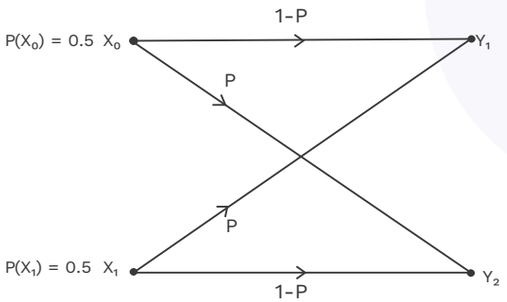
= Max H(x) = Max H(y)

$= \log_2 N = \log_2 M$

N = M

i.e. for noiseless channel, number of input symbols = number of output symbols.

Channel capacity of binary symmetric channel:



Now, from the above channel,

$$P(y_0; x_0) = P\left(\frac{y_o}{x_o}\right)P(x_o)=0.5(1-p)$$

$$P(y_1,x_1)=P\left(\frac{y_1}{x_1}\right)P(x_1)=0.5(1-P)$$

$$P(y_0,x_1)=P\left(\frac{y_0}{x_1}\right)P(x_1)=p\times 0.5$$

$$P(y_1,x_0)=P\left(\frac{y_1}{x_0}\right)P(x_0)=p\times 0.5$$

$$H\left(\frac{y}{x}\right)=\sum_{i=0}^{N-1}\sum_{j=0}^{K-1}P(x_i,y_j)\log_2\frac{1}{P(y_j/x_i)}$$

$$=0.5(1-P)\log_2\left(\frac{1}{1-P}\right)+0.5(1-P)\log_2\frac{1}{(1-P_i)}$$

$$+0.5\log_2\frac{1}{P}+0.5\log_2\frac{1}{P}$$

$$H\left(\frac{y}{x}\right)=-(1-P)\log_2(1-P)-P\log_2 P$$

$$I=H(y)-H\left(\frac{y}{x}\right)$$

Max(I) = 1 + P $\log_2$P + (1 – P)$\log_2$(1 – P)

Capacity = 1 + P $\log_2$P + (1 – P)$\log_2$(1 – P)

Source Coding & Source Coding Theorem



Source coding:

The source produces the symbols, and the source coding is done to provide digital code to each symbol. Here, the motive of the source coding is to perform lossless data compaction. For example, if a source produces 4 symbols - S_0, S_1, S_2, and S_3 with probabilities 0.5, 0.3, 0.1 and 0.1 in a respective manner. Now, the average length of the code without source coding

Code

S_0→ 0.5 00

S_1 → 0.3 01

S_2→ 0.1 10

S_3→ 0.1 11

Average length of code = $\sum_{i=o}^{i=n-1} L_i P_i$

Where L_i = Length of code for i^{th} symbol and P_i = Probability of ith symbol

n = Number of codes

Now,

Average code length

= 2 × 0.5 + 2 × 0.3 + 2 × 0.1 + 2 × 0.1

Average code length = 2 bits/symbol

Now,

Average code length with source coding probability code

Code

S_0→ 0.5 1

$S_1 \rightarrow 0.3 \quad 0$

$S_2 \rightarrow 0.1 \quad 11$

$S_3 \rightarrow 0.1 \quad 10$

Now,

Average code length

$= 0.5 \times 1 + 0.3 \times 1 + 0.1 \times 2 + 0.1 \times 2$

$= 1.2$ bits/symbol

So, from the example above, we can say that if we assign small code-words to the symbols that have a high probability, leads to data compaction, which is lossless. And this is done by using a source encoder.

Source coding theorem:

The theorem states that average code-word length

$$i.e. \left[L_{avg} = \sum \Pr obability\ of\ symbol \times codeword\ length\ of\ symbol \right]$$

is always greater than or equal to entropy of the source

i.e. L avg ≥ H(x)

This theorem is also known as Shannon's 1st theorem. Efficiency of source coder –

$$= \frac{Minimum\ codeword\ length\ possible}{Average\ codeword\ length\ produced} \times 100$$

$$\text{Code redundancy} = \left(1 - \frac{H(x)}{L_{avg}} \right)$$

Code redundancy = $(1 - \eta)$

$$\text{Efficiency} = \frac{H(x)}{L_{avg}} \times 100$$

Methods of Source Encoding

Huffman coding:

Huffman encoding is a lossless data compaction algorithm. The main idea is to assign small code-words to high probability symbols and large code words to low probability symbols

Procedure:

1. Arrange all symbols in decreasing order of probability.
2. Combine the probabilities of last 2 symbols and again rearrange the resultant probabilities and repeat step 1 and step 2 till only 2 symbols are left.
3. Now assign bit 1 0 or 0 1 to each pair and find out codes to each symbol.

Example:

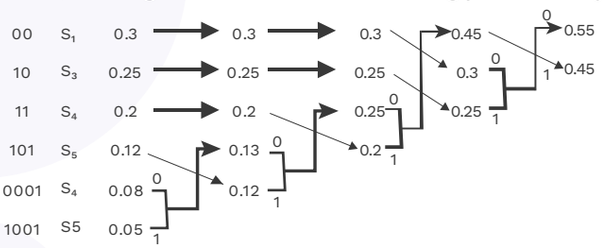
A source emits 6 symbols, S_1, S_2, S_3, S_4, S_5 and S_6 with probabilities 0.3, 0.2, 0.25, 0.08, 0.12 and 0.05 respectively. If we are using Huffman coding, then find the efficiency of source coder.

Solution:

Given,

Symbols	S1	S2	S3	S4	S5	S6
Prob-ability	0.3	0.2	0.25	0.08	0.12	0.05

Now, arrange them with decreasing probability



Symbols	Prob-ability	Codeword	Codeword Length
S1	0.3	00	2
S2 0.2	11	2	2
S3	0.25	10	2
S4	0.08	0001	4
S5	0.12	101	3
S6	0.05	1001	4

Average code word length

$$= \sum_{i=1}^{i=6} L_i P_i$$

$$L_{avg} = 0.3 \times 2 + 0.25 \times 2 + 0.2 \times 2 + 0.08 \times 4 + 0.12 \times 3 + 0.05 \times 4$$

= 0.6 + 0.4 + 0.5 + 0.32 + 0.36 + 0.2

L_{avg} = 2.38 bits/symbol

$$Efficiency\ \eta = \frac{H(x)}{L_{avg}} \times 100$$

$$H(X) = \sum P(X_i) \log \frac{1}{P(X_i)}$$

$$= 0.3 \log_2 \frac{1}{0.3} + 0.2 \log_2 \frac{1}{0.2} + 0.25 \log_2 \frac{1}{0.25}$$

$$+ 0.08 \log_2 \frac{1}{0.08} + 0.12 \log_2 \frac{1}{0.12} + 0.05 \log_2 \frac{1}{0.05}$$

= 0.5211 + 0.4643 + 0.5 + 0.2915 + 0.3671 + 0.2161 = 2.3601

H(x) = 2.36

L_{avg} = 2.38

Shannon-Fano coding:

Procedure:

1. Arrange the symbols in decreasing order of probabilities.
2. Divides the symbols into two sets with the probability of sets close to ½, as possible, then, those sets are themselves divide into two parts, keeping the probability of sets near to half of the previous set. This process is done until each set contains only one symbol, then code the sets from 1st two sets as 1 and 0. e.g. if a source produces 4 symbols with probabilities 0.3, 0.2, 0.25 and 0.25. Arrange them in decreasing order

Symbols	Probability		Shannon - Fano coding
S_1	0.3	1	1
S_3	0.25	1	0
S_2	0.25	0	1
S_4	0.2	0	0

Code word for different symbols

Symbols	Probability	Codeword
S1	0.3	11
S2	0.2	00
S3	0.25	10
S4	0.25	01

Example:

A source emits 6 symbols, S_1, S_2, S_3, S_4, S_5 and S_6 with probabilities 0.3, 0.2, 0.25, 0.08, 0.12 and 0.05 respectively. If we are using Shannon-Fano coding, find the efficiency of source code and amount of redundancy with respect to L_{avg}.

Solution:

Given:

Symbols	51	52	53	54	55	56
Prob-ability	0.3	0.2	0.25	0.08	0.12	0.05

Shannon-Fano coding

S1	0.3	1	1			
S2	0.25	1	0			
S3	0.2	0	1			
S4	0.12	0	0	1		
S5	0.08	0	0	0	1	
S6	0.05	0	0	0	0	

Code-word and code-word length

Symbols	Prob-ability	Codeword	Codeword Length
S1	0.3	11	2
S2	0.2	01	2
S3	0.25	10	2
S4	0.08	0001	4
S5	0.12	001	3
S6	0.05	0000	4

$$L_{avg} = \sum_{i=1}^{i=6} L_i P_i$$

= 2 ×0.3 + 2 × 0.25 + 2 ×0.2 + 4 ×0.08 + 3 ×0.12 + 4 ×0.05

= 0.6 + 0.25 + 0.4 + 0.32 + 0.36 + 0.2 = 2.38

$$H(x)=\sum P(x_i)\log_2\frac{1}{P(x_i)}=2.36$$

$$Efficiency\ \eta=\frac{H(x)}{L_{avg}}\times 100$$

$$Efficiency\ '\eta'=\frac{2.36}{2.38}\times 100$$

= 99.15%

Redundancy = 1 – η

= 1 – 0.9915

Redundancy = 0.0085

Classification of Codes

1. **Prefix-free codes:** A code is called a prefix-free code if no code-word is a prefix of another one. These are also known as prefix codes.

Symbols	Code (1)	Code (2)	Code (3)
S1	00	0	0
S2	01	1	10
S3	10	00	110
S4	11	11	111
	↓	↓	↓
	Prefix - free code	Non - prefix free code	Prefix - free code

2. **Uniquely decodable codes:** If the code-word is defined properly, the receiver can detect/decode the code-word uniquely. All prefix-free codes are uniquely decodable, since, no other code-word is prefix of another code-word, so, there is no confusion at receiver.
3. **Instantaneous codes:** The codes which can be decoded instantaneously at the receiver are called instantaneous codes. All uniquely decodable codes are instantaneous codes.
4. **Optimal codes:** If the code-words are instantaneous and are of minimum lengths, then, they are optimal codes.

   Kraft's inequality: It states that if l_K is the length of code of Kth symbols.

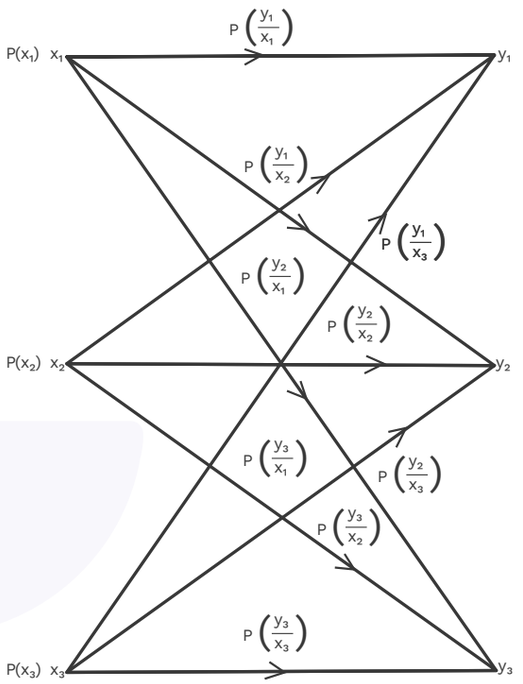
   Then,

   $$\sum_{k=1}^{N}2^{-l_k}\le 1$$

N = Number of symbol in Kraft's inequality is necessary and sufficient condition for a code to be instantaneous code.

Optimum Detection (MAP)

$P(x_2)$



⇒ if y_k is detected as x_m

then,

$$P(x_m)P\left(\frac{y_k}{x_m}\right)\rangle P(x_l)P\left(\frac{y_k}{x_l}\right)\ m\neq l$$

If, y_k is detected as x_m half of the time and x_n half of the time

Then, $P(x_m)P\left(\frac{y_k}{x_m}\right)\rangle P(x_n)P\left(\frac{y_k}{x_n}\right)$

For example,

⇒ if y_1 detected as x_1

y_2 detected as x_2

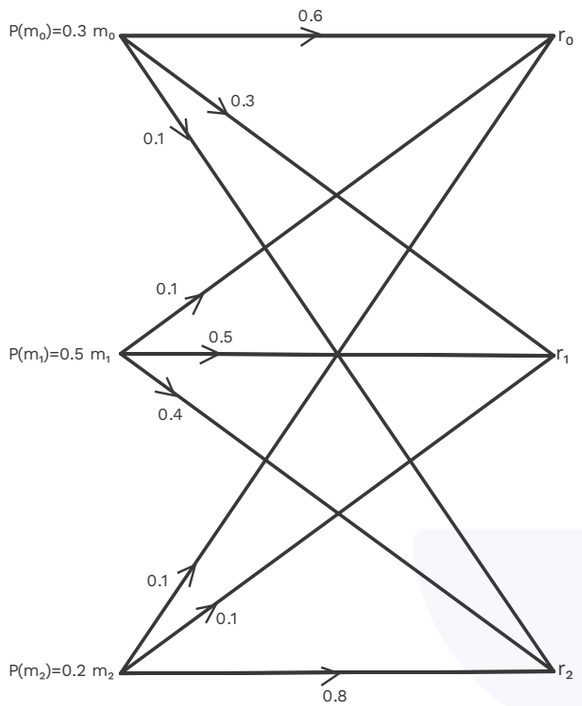
y_3 detected as x_3

$$P_c=P(x_2)P\left(\frac{y_1}{x_2}\right)+P(x_2)P\left(\frac{y_2}{x_2}\right)+P(x_3)P\left(\frac{y_3}{x_3}\right)$$

Probability of error, $P_e = 1-P_c$

Example:

For a channel given in the figure below, for each possible received response with decision in the above, calculate the probability of error.



Best decision:

	m_0	m_1	m_2	Decisions
r_0	0.18	0.05	0.02	r_0 is detected as m_0
r_1	0.09	0.25	0.02	r_1 is detected as m_1
r_2	0.03	0.20	0.16	r_2 is detected as m_1

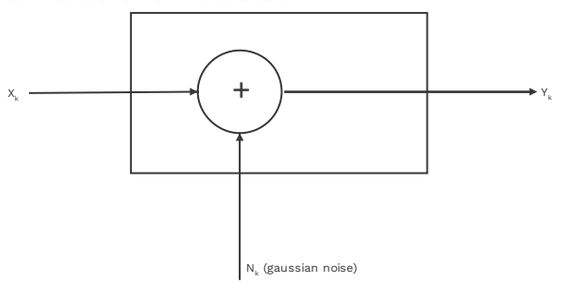
P_c = 0.18 + 0.25 + 0.20

P_c = 0.63

Probability of error, P_e = 0.37

Information Capacity Theorem (for AWGN Channel)

Consider a Gaussian channel band-limited to B Hz as shown below



The source transmits the symbol X_k and noise N_k gets added in it and we

get Y_k at output.

i.e. $Y_k = X_k + N_k$

Where, N_k is the components of gaussian noise. If, B is the bandwidth of message, then the variance of noise = N_0B y. Let, P is the power of message being transmitted.

Channel capacity = Max[I(X : Y)

$$I(X ; Y) = H(Y) - H\left(\frac{Y}{X}\right)$$

Now, since $Y_k = X_k + N_k$ y If, noise ≠ 0

$H\left(\frac{Y}{X}\right) \neq 0$ [uncertainty in output given that input is known. (This is only due to noise in the channel)]

If, noise = 0, i.e., noiseless channel

Then,

$$H\left(\frac{Y}{X}\right) = 0$$

But, here, channel has noise

Then,

$$H\left(\frac{Y}{X}\right) = \textit{entropy of noise}$$

Since, noise is Gaussian,

So,

$$H\left(\frac{Y}{X}\right) = H(N_k) = \frac{1}{2}\log_2\left(2\sigma^2 e\right)$$

As max[I(X ; Y)] = Channel capacity

Therefore,

To get max mutual information, we need H(Y) to be maximum. Here, Y_k needs to be gaussian, so, we will take X_k to be Gaussian for max mutual information.

$$\underbrace{Y_k}_{\text{gaussian}} = \underbrace{X_k}_{\text{gaussian}} + \underbrace{N_k}_{\text{gaussian}}$$

$$H(y_k) = \frac{1}{2}\log_2\left(2\pi\sigma_y^2\, e\right)$$

$$\sigma_y^2 = P + N_0 B$$

Now,

Channel capacity = Max information

$$= \frac{1}{2}\log_2\left[2\pi(P + N_0B)e - \frac{1}{2}\log_2(2\pi N_0Be)\right]$$

$$C = \frac{1}{2}\log_2\left(1 + \frac{P}{N_0B}\right) \text{ Bits / symbol}$$

Here, B is the bandwidth of message, so, we need to send 2B symbol/sec.

The following probabilities are given

$$C = 2B \times \frac{1}{2}\log_2\left(1 + \frac{P}{N_0B}\right) \text{ Bits / sec}$$

$$C = B\log_2\left(1 + \frac{P}{N_o}\right) \text{ Bits / sec}$$

$$C = B\log_2(1 + SNR) \text{ Bits / sec}$$

Now, if 'B' is infinite

$$\lim_{B\to\infty} C = \log_2\left(1 + \frac{P_x}{N_0B}\right)^B$$

$$= \log_2 e^{P_x/N_o}$$

$$C = \frac{P_x}{N_0}\log_2 e = 1.44\frac{P_x}{N_o}$$

P_x = Average signal power

Implication of Information Capacity Theorem

We know that for correct working, the bit rate, $R_b \le$ Channel capacity, C

i.e., $R_b \le C$

Let us take the critical condition that $R_b = C$

Now,

$$C = B\log_2\left(1 + \frac{P}{N_0B}\right)$$

$$P = E_b \times R_b = E_b \times C$$

$$R_b \le B\log_2\left(1 + \frac{E_bR_b}{N_0B}\right)$$

$$\frac{R_b}{B} < \log_2\left(1 + \frac{E_bR_b}{N_0B}\right)$$

$$2^{R_b/B} \le 1 + \frac{E_bR_b}{N_0B}$$

For error free T_x

$$\frac{E_b}{N_0} = \frac{2^{R_b/B} - 1}{\left(\frac{R_b}{B}\right)}$$

For infinite Bandwidth channel

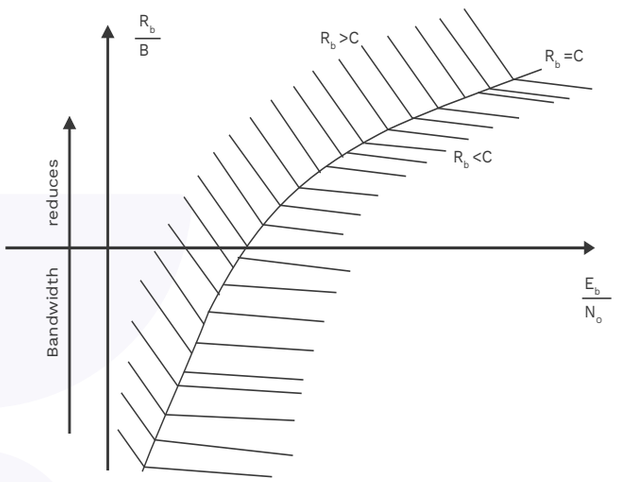
If $= B \to \infty \to \frac{R_b}{B} \to 0$

$$= \lim_{\frac{R_b}{B}\to 0} = \frac{2^{R_b/B}}{\frac{R_b}{B}} = \ln 2$$

$$\frac{E_b}{N_o} \ge \ln 2$$

$$\frac{E_b}{N_o} \ge 0.693$$

$$\left(\frac{E_b}{N_0}\right) \ge -1.59dB$$



Error Control Technique

For the correction of errors, we must detect the error prior to correction. Send ARQ (Automatic Repeat Request): On detecting the error, the receiver will request the transmitter to resend the data.

Linear Block Codes (n, k)

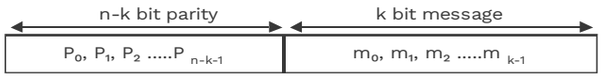
The parity bits of linear block codes are linear combinations of the message. Linear block code is a type of error-correcting code in which the actual information bits are linearly combined with the parity check bits. If, k bit message is encoded in n bit code-word,

Codeword = n bits

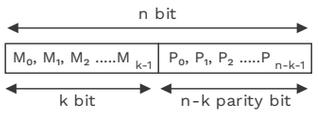
Message = k bits

Parity bit = n – k bits

In linear block code,



We can also make the arrangement as follows



Here, parity bits are created using message bits only. For example,

$P_0 = b_{00}m_0 + b_{01}m_1 + b_{02}m_2 + b_{03}m_3 + \ldots b_{k-1}m_{k-1}$

$P_1 = b_{10}m_0 + b_{11}m_1 + b_{12}m_2 \ldots b_{1k-1}m_{k-1}$

Let us say, n = 7, k = 4

So, parity bits = n – k = 7 – 4 = 3 bits

Parity matrix = $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$

Parity bit = $\underbrace{\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}}_{\text{Parity matrix}} \underbrace{\begin{bmatrix} m_0 \\ m_1 \\ m_2 \\ m_3 \end{bmatrix}}_{\text{Message vector}}$

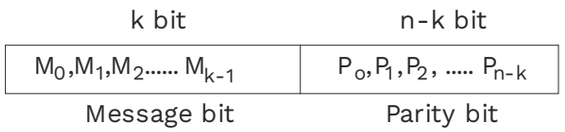
Let message vector is = [1 1 0 1]

Now parity bit,

$$P = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 \oplus 1 \oplus 1 \\ 1 \oplus 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Now code-word:



Code-word = $\underbrace{1101}_{\text{Message bit (4 bit)}}\ \underbrace{101}_{\text{Parity Big (3 bit)}}$

Now, the parity matrix,

$$B = \begin{bmatrix} b_{oo} & b_{01} & \ldots\ldots b_{ok-1} \\ b_{10} & b_{11} & \ldots\ldots b_{1k-1} \\ . & . & . \\ . & . & . \\ . & . & . \\ b_{n-k0} & b_{n-k1} & \ldots\ldots b_{n-k,k-1} \end{bmatrix}_{(n-k)\times k}$$

Parity vector,

$$P = \begin{bmatrix} B \end{bmatrix}_{(n-k)\times k} \begin{bmatrix} Message \\ Vector \end{bmatrix}_{k\times 1}$$

Massage vector $= \begin{bmatrix} Message \\ Vector \end{bmatrix}_{1\times k} \begin{bmatrix} I \end{bmatrix}_{k\times k}$

Code-word = [Message: Parity]

Code-word =

$$\begin{bmatrix} m_0, m_1, m_2, \ldots\ldots m_{k-1} \end{bmatrix} \underbrace{\left[(I)_{k\times k} : \begin{bmatrix} Parity \\ matrix \end{bmatrix} \right]}_{\text{Generator matrix}}$$

$$\text{Generator matrix } G = \left[(I)_{k\times k} : \begin{bmatrix} Parity \\ matrix \end{bmatrix}_{k\times b-k} \right]$$

$$\begin{bmatrix} Code\ word \end{bmatrix}_{1\times n} = \begin{bmatrix} Message\ vector \end{bmatrix}_{1\times k} \begin{bmatrix} G \end{bmatrix}_{k\times n}$$

[Code-word] = [Message vector] [G]

The size of generator matrix is (k × n), it has 2 parts - identity matrix and parity matrix.

Identity matrix is of dimension k × k;

Parity matrix is of dimension k × n – k;

Why these codes are known as linear block code?

Let us suppose, message is m_j and its code $c_j = m_j G$ and let another message be m_i and its code $c_i = m_i G$

Now, $m_j \oplus m_i$ (Bit wise EX-OR) will be another message signal. So, we can say that if we add two code words $c_j + c_i = (m_j + m_i) G$; i.e., Sum of 2 code-words = another code-word. As, here, if we add 2 code-words, we will get a new code-word and due to these, code-words are known as linear code words.

The addition here is not binary addition, it implies bitwise EX-OR or module-2 addition.

eg. 10110+11001	01111
(*Assition of two codeword*)	(*New codeword*)

Now, the code-word is transmitted and at the receiver, the code-word is checked by the check matrix, also called as parity check matrix 'H'.

Parity check matrix:

Parity check matrix:

$$'H' = \left[\left[B \right]^{T}_{n-k \times k} \vdots \left[I \right]_{n-k \times n-k} \right]$$

$$H^{T} = \begin{bmatrix} \left[B \right]_{k \times n-k} \\ \cdots\cdots\cdots\cdots \\ \left[I \right]_{n-k \times n-k} \end{bmatrix}_{n \times n-k}$$

As we know that, C = mG

$$C = mG = m\left[\left[I \right]_{k \times k} \vdots \left[B \right]_{k \times n-k} \right]$$

Then-

$$CH^{T} = \left[\left[I \right]_{k \times k} \vdots \left[B \right]_{k \times n-k} \right] \begin{bmatrix} \left[B \right]_{k \times n-k} \\ \cdots\cdots\cdots\cdots \\ \left[I \right]_{n-k \times n-k} \end{bmatrix} = \left[0 \right]$$

i.e.

$$CH^{T} = \left[0 \right] \left[As\ these\ are\ module-2\ addition \right]$$

eg: If,

$$C = m \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}_{2 \times 5}$$

$$H^{T} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{5 \times 3}$$

$$CH^{T} = m \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}_{2 \times 5} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{5 \times 3}$$

$$= \begin{bmatrix} 1 \oplus 1 & 1 \oplus 1 & 0 \\ 1 \oplus 1 & 0 & 1 \oplus 1 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$CH^{T} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2 \times 3}$$

As we are getting $CH^{T} = 0$, it implies that the received code-word is same as transmitted code word, i.e., there is no error in transmission.

Example:

Given, a generator matrix G. Find the parity matrix, code rate and parity. For the given generator matrix –

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}_{3 \times 6}$$

Solution:

Given, generator matrix G

And, we know that

Generator matrix

$$G = \left[\left[I \right]_{k \times k} \vdots \left[B \right]_{k \times n-k} \right]$$

$$\left[B \right]_{k \times n-k} = Parity\ matrix$$

Generator matrix $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$ (I | B)

From the above given generator matrix:

$$\left[I \right] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

And, parity matrix:

$$\left[B \right] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

As the size of generator matrix is k × n = 3 × 6

Hence, k = 3, n = 6

Number of parities = n – k

= 6 – 3

= 3

Code rate = $\frac{k}{n} = \frac{3}{6} = 0.5$

Parity check matrix $H = \left[\left[B \right]^{T}_{n-k \times k} \vdots \left[I \right]_{k \times k} \right]$

$$H = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] = \left[\underbrace{\begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}}_{(B)^T} \; \underbrace{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}}_{(I)} \right]$$

What happens if the code word has error?

If received code word has error vector 'e'.

r = c + e

Received codeword — Ccodeword — Error vector

If, error vector = All elements zero, then, no error

If, error vector = Some non-zero elements, then there is an error in received vector. e.g. If we are transmitting code-word = 0011 and we receive code-word as = 0001; as, received codeword is different from the transmitted code-word, i.e., there is an error in the received code-word. The element of error vector 'e' is 1 for the bit we have received incorrectly. As, here, 3rd bit is incorrectly received, then the error vector = 0010 (3rd place bit = 1) or in other words, if any bit of the error vector is 1, then, there is an error in the received code-word on the same bit.

Now, as received code-word r = C + e

$$rH^T = (C + e)H^T$$
$$= CH^T + eH^T$$

and $CH^T = 0$

$$rH^T = eH^T$$

Case-1:

If $rH^T = eH^T \neq 0$, error in received code-word.

Case-2:

$rH^T = eH^T = 0$, no error in received code-word.

If, eH^T = any row of H^T, it means that code-word has 1 bit error. If, $eH^T = k^{th}$ row of H^T, then, k^{th} bit of received code-word has error.

If, received code-word has 2 bit error hence, error vector has '1' in 2 locations and eH^T = Sum of 2 rows of H^T. If, the received code-word has 3 bit error, then the error vector has '1' in 3 locations and eH^T = Sum of 3 rows of H^T.

Hamming Distance and Hamming Weight

If, we perform bit wise EX-OR of 2 codes then number of 1's result is Hamming distance

e.g. code 1 = 1 1 1 0 1

code 2 = 0 1 0 1 0

EX-OR of code 1 and code 2

1 1 1 0 1
0 1 0 1 0
_____
1 0 1 1 1

Number of 1 in bit wise EX-OR is 4

Hence, Hamming distance = 4

Hamming weight:

Hamming weight of a code is Hamming distance of a code from all '0' code or number of '1' in any code word is Hamming weight.

e.g. code-word = 1 1 0 1 0 1

Hamming distance from all zero code

1 1 0 1 0 1
0 0 0 0 0 0
_____
1 1 0 1 0 1

Number of '1' = 4 = Hamming weight

Example:

Consider 2 code-words C_1 and C_2. Calculate the Hamming weight of both code and Hamming distance between codes.

C_1 = 01010101

C_2 = 11100010

Solution:

Hamming weight of code C_1 = Number of '1' in code C_1

Hamming weight of code C_1 = 4

Hamming weight of code C_2 = Number of '1' in codes C_2

Hamming weight of code C_2 = 4

Hamming distance between code C_1 and C_2 = number of '1' in bit wise EX-OR of both code

0 1 0 1 0 1 0 1

1 1 1 0 0 0 1 0

---

1 0 1 1 0 1 1 1

Hamming distance = Number of '1' in bitwise EX-OR

= 6

Proper code-word:

In (n, k) linear block code

Number of proper code words = 2k

Proper code word follows the given inequality equation

$$= \sum_{i=0}^{t} 2^k \, {}^nc_i \le 2^n$$

$$= 2^k \sum_{i=0}^{t} {}^nc_i \le 2^n$$

$$\sum_{i=0}^{t} {}^nc_i \le 2^{n-k} \text{ –Hamming Bound}$$

⇒ If any code-word system (techniques), which holds equality, then, coding is called the perfect code.

⇒ Hamming code is a perfect code with single bit error correction. Therefore

$$\left({}^nc_0 + {}^nc_1\right) = 2^{n-k}$$

$$1 + n = 2^{n-k}$$

$$n = 2^{n-k} - 1$$

$$n - k = m$$

$$k = n - m$$

$$k = 2^{n-k} - 1 - m$$

$$\boxed{k = 2^m - 1 - m}$$

Hamming code $(2^m - 1, 2^m - 1 - m)$

⇒ Minimum distance of a code-word set

$d_{min} < n\text{-}k + 1$

⇒ Coding rate for (n, k) linear block code $R = \frac{k}{n}$

Exercise

1. An analogue baseband signal band, limited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four message symbols that occur independently with probabilities $P_1 = P_4 = 0.125$ and $P_2 = P_3$. The information rate (bits/sec) of the message source is _____.

Solution: 362.2

2. A voice–grade A WGN (Additive White Gaussian Noise) telephone channel has a bandwidth of 4.0 kHz and two – sided noise power spectral density

$\frac{\eta}{2} = 2.5 \times 10^{-5}$ Watt per Hz

If, the information at the rate of 52 kbps is to be transmitted over this channel with an arbitrarily small bit error rate, then the minimum bit – energy E_b (in mJ/bit) necessary is _____.

Solution: 31.5

3. Consider a binary channel code, in which, each code-word has a fixed length of 5 bits. The Hamming distance between any pair of distinct code-words in this code is at least 2. The maximum number of code-words such a code can contain is _____.

Solution: 16

4. A linear hamming code is used to map 4 bit message to 7 bit code-word. The encoder mapping is linear. If, message 0001 is mapping to the code-word 0000111 and the message 0011 is mapped to code-word 1100110, then, message 0010 is mapped to

(A) 1100001

(B) 0010011

(C) 1111000

(D) 1111111

Solution: A

5. A digital transmission system uses a (7, 4) systematic linear Hamming code for transmitting data over a noisy channel. If, three of the messages – codeword pairs in this code $(m_i; c_i)$, where, c_i is the code-word corresponding to the ith message m_i, are known to be (1100; 0101100), (1110; 0011110) and (0110; 1000110), then, which of the following is a valid code-word in this code?

(A) 1101001
(B) 1011010
(C) 0001011
(D) 0110100

Solution: C

6. A binary random variable X takes the value +2 or −2. The probability $P(X = +2) = \alpha$. The value of α (rounded off to one decimal place), for which the entropy of X is maximum, is _____.

Solution: 0.5

Solutions

1. Given:

$f_m = 100\,\text{Hz}$

and $f_s = 2f_m = 200$ sample/sec $= r$

$P_1 = P_4 = \frac{1}{8}$ and $P_2 = P_3$

Since, $P_1 + P_2 + P_3 + P_4 = 1$

$0.125 + P_2 + P_2 + 0.125 = 1$

$P_2 = \frac{3}{8}$

So, $P_2 = P_3 = \frac{3}{8}$

Entropy of the source X is given by,

$$H = \sum_{i=1}^{N} P_i \log_2 \frac{1}{P_i}$$

$$H = \frac{1}{8}\log_2 8 + \frac{1}{8}\log_2 8 + \frac{3}{8}\log_2 \frac{8}{3} + \frac{3}{8}\log_2 \frac{8}{3}$$

$H = 1.811$ bits/sample

Information rate is given by,

$R = rH$

$R = 1.811 \times 200 = 362.2\,\text{bits/sec}$

Hence, the information rate of the message source is 362.2 bits/sec.

2. Given:

Power spectral density, $\frac{\eta}{2} = 2.5 \times 10^{-5}$

and information rate $= 52$ kbps

Capacity of the channel is given by,

$$C = B\log_2\left(1 + \frac{S}{N}\right)$$

where, S = signal power, N = Noise power, B = Bandwidth.

$S = E_b \cdot R_b$

Where, E_b = Bit energy and R_b = Information rate bits/sec.

$$C = B\log_2\left(1 + \frac{E_b R_b}{\eta B}\right)$$

For distortion less transmission, C should be at least equal to R_b.

$C \geq R_b$

$$R_b = B\log_2\left[1 + \frac{E_b R_b}{\eta B}\right]$$

$$\frac{R_b}{B} = \log_2\left[1 + \frac{E_b R_b}{\eta B}\right]$$

$$E_b = \left[2^{\frac{R_b}{B}} - 1\right]\frac{\eta B}{R_b}$$

$$E_b = \frac{\left[2^{\frac{52}{4}} - 1\right] \times 2 \times 2.5 \times 10^{-5} \times 4}{52}$$

$E_b \approx 31.504$ mJ/bit

Hence, the minimum bit – energy E_b necessary is 31.504 mJ/bit.

3. Given:

Fixed length of code $n = 5$

Minimum hamming distance $d_{mis} = 2$.

For hamming code $(n, k) d_{\min}$ is given by

$d_{\min} \geq n - k + 1$

$2 \geq 5 - k + 1$

$k \geq 6 - 2 \geq 4$

All the possible code words having minimum Hamming distance $d_{\min} = 2$

$2^k = 2^4 = 16$

C_4	C_3	C_2	C_1	C_0
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Hence, the maximum number of code words such a code can contain is 16.

4. Given:

Data				Codeword						
0	0	0	1	0	0	0	0	1	1	1
0	0	1	1	1	1	0	0	1	1	0

Linear Hamming code i.e. mod 2 addition of any two codewords will be codewords of code vector

	Data				Codeword						
Mod 2 $\oplus$	0	0	0	1	0	0	0	0	1	1	1
	0	0	1	1	1	1	0	0	1	1	0
	0	0	1	0	1	1	0	0	0	0	1

Hence, the correct option is (A).

5. Given:

Message and corresponding codewords are,

Message				Codeword						
1	1	0	0	0	1	0	1	1	0	0
1	1	1	0	0	0	1	1	1	1	0
0	1	1	0	1	0	0	0	1	1	0

From below four parity generator equation we have to select any three,

$P_A = m_1 \oplus m_2 \oplus m_3$

$P_B = m_1 \oplus m_2 \oplus m_4$

$P_C = m_2 \oplus m_3 \oplus m_4$

$P_D = m_1 \oplus m_3 \oplus m_4$

Form comparing with all three given codewords i.e., C_{12}, C_{14} and C_6

$P_1 = P_B, P_2 = P_C, P_3 = P_4$

So, for message $m_9 = 1001, C_9 = 0111001$

Option (A) is wrong.

$m_{10} = 1010,\ C_{10} = 1101010$

Option (B) is wrong.

$m_{11} = 1011, C_{11} = 0001011$

Option (C) is correct.

$m_4 = 0100, C_4 = 1110100$

Option (D) is wrong.

Hence, the correct option is (C).

10. Given:

A random variable 'X' takes value +2 or -2.

Since, $P(X=+2)=\alpha$

Then $P(X=-2)=1-\alpha$

Entropy is given by,

$$H=\sum_i P(x_i)\log_2\frac{1}{P(x_i)}$$

$$H=\alpha\log_2\frac{1}{\alpha}+(1-\alpha)\log_2\left(\frac{1}{1-\alpha}\right)$$

$$H=-\alpha\log_2\alpha+(\alpha-1)\log_2(1-\alpha)$$

For maxima or minima,

$$\frac{dH}{d\alpha}=0$$

$$\frac{d}{d\alpha}[(\alpha-1)\log_2(1-\alpha)-\alpha\log_2\alpha]=0$$

$$\frac{1}{\log_e 2}\frac{d}{d\alpha}[(\alpha-1)\log_e(1-\alpha)-\alpha\log_e\alpha]=0$$

$$\left[(\alpha-1)\times\frac{1}{1-\alpha}\times(-1)+\log_e(1-\alpha)(1-0)\right]$$

$$-\left(\alpha\frac{1}{\alpha}+\log_e\alpha\right)=0$$

$$1+\log_e(1-\alpha)-1-\log_e\alpha=0$$

$$\log_e(1-\alpha)=\log_e\alpha$$

$$1-\alpha=\alpha$$

$$2\alpha=1$$

$$\alpha=\frac{1}{2}=0.5$$

Hence, the entropy of X will be maximum when α is 0.5.